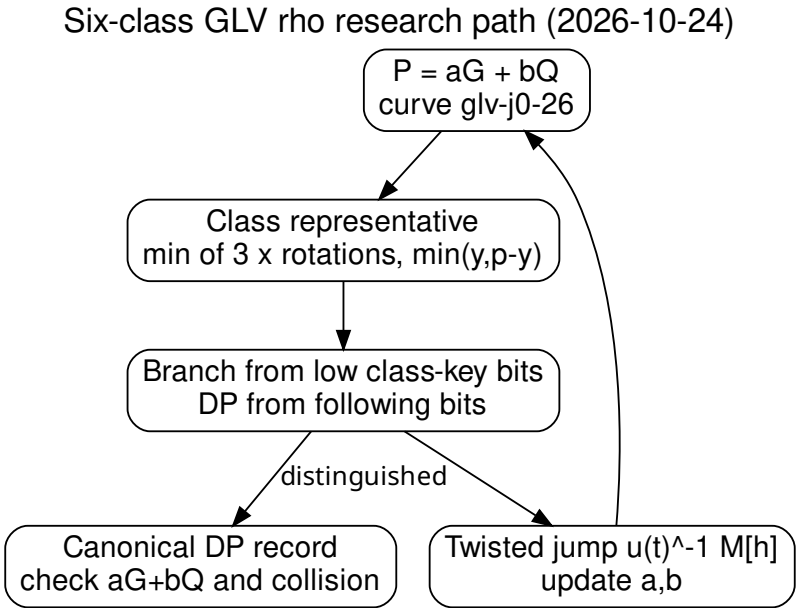
Six-class GLV rho assessment

2026-10-24

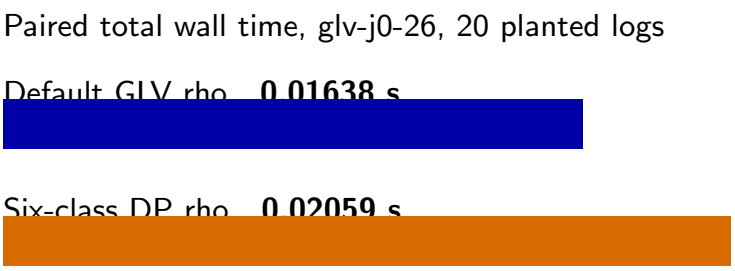
Question and method

We assessed the six-class GLV walk from Curve Smash commit 1f084c1 against our native default on registered curve `glv-j0-26`. Its class representative minimizes the three x rotations and the sign of y. The orientation twists the jump, and distinguished point records carry canonical coefficients.



Measured result

In one paired local run, both solvers recovered all 20 planted logs. Default oriented GLV rho took 0.016382582 s; six-class DP rho took 0.020590401 s, 1.257 times as long. The candidate visited 46,882 classes, emitted 2,922 DPs, and took 196 short-cycle escapes. These are total times, with no uncertainty estimate from this single run.



Interpretation and limits

This is a negative toy-curve result. It is not a secp256k1 cost measurement. Commit 1f084c1 changes κ from 1.05 to 1.02 and the inversion batch to 2^{16} while leaving the algorithm unchanged. Its 126.96-bit cost and ≥ 0.3909 success remain conditional on the stated H1 mixing and cost model. The C solver is opt-in. A Rust BigUint worker, harness launcher and native Cairn DP checker provide an end-to-end research path. The C and Rust job derivations are different, so their receipts cannot be pooled. DP point equations do not prove walker provenance; sampled re-walks are still required.

Evidence and next checks

The source record is `assessment.md`; raw receipt is `paired-glv-j0-26.json`. Repeat paired A/A and A/B tests over the curve ladder, audit sampled trails, then assess a fixed-limb implementation. No isogeny-route or volcano graph update follows from this walk.